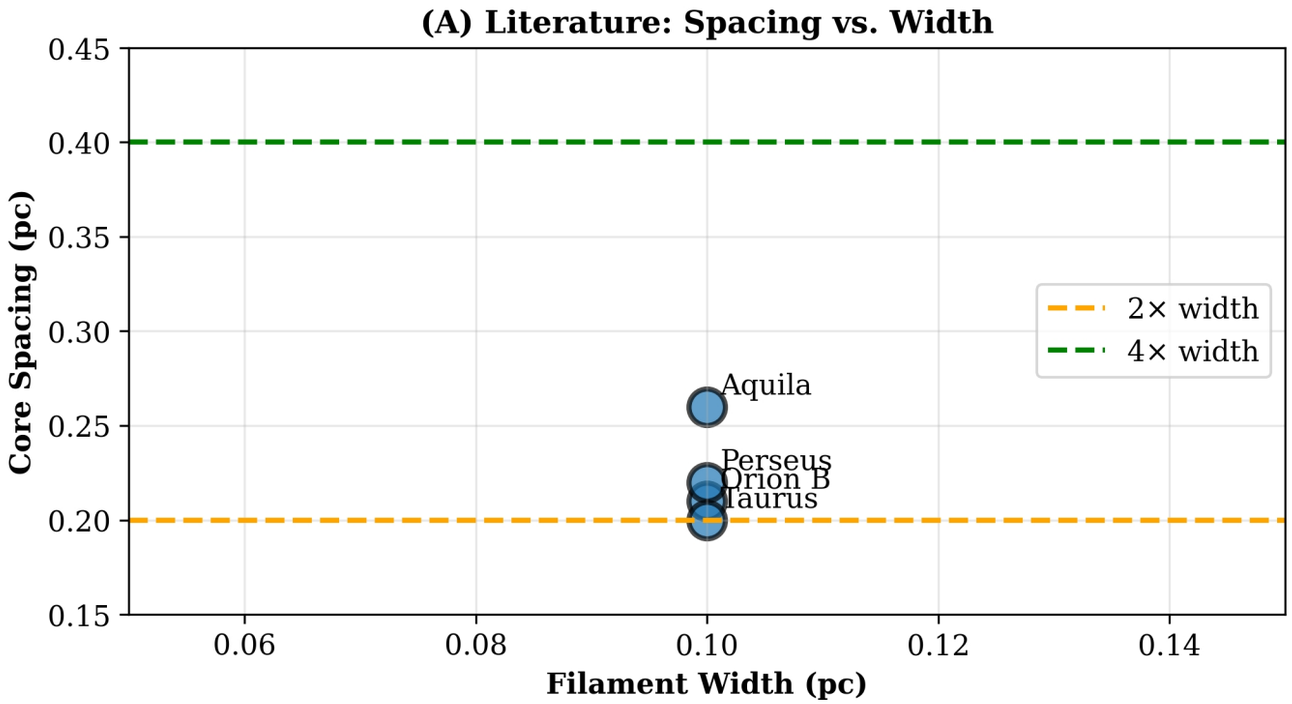
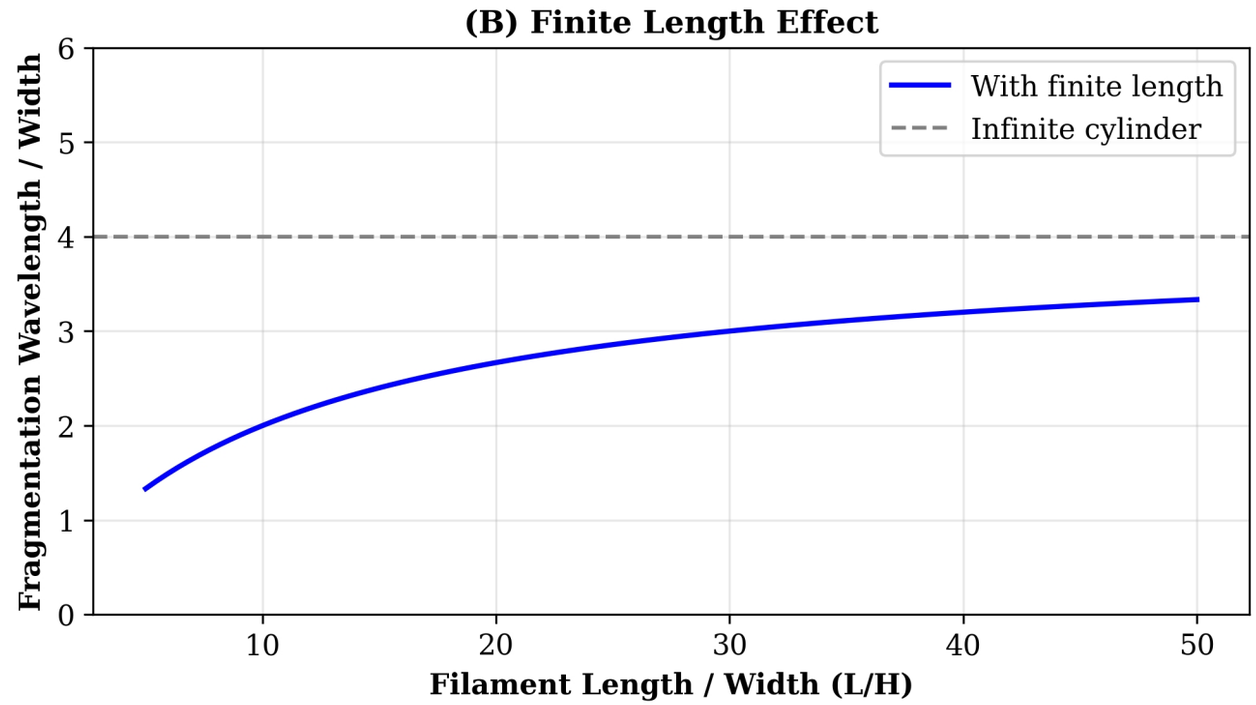
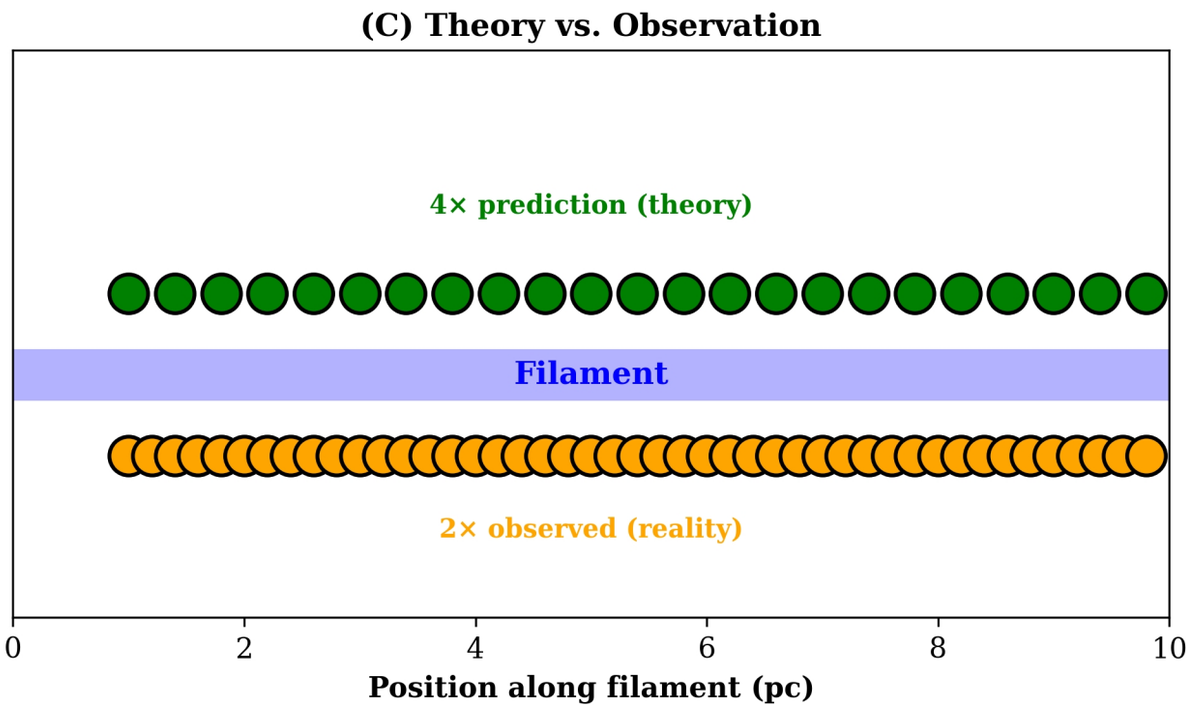
Investigation of 2× vs. 4× Filament Spacing Discrepancy





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WHY IS OBSERVED SPACING ~2× NOT 4×?

1. FINITE LENGTH EFFECTS
   Shorter filaments fragment at
   shorter wavelengths

2. EXTERNAL PRESSURE
   Surrounding gas compresses
   filaments, changes λ_max

3. TAPERED GEOMETRY
   Real filaments are not perfect
   cylinders - they narrow

4. MASS ACCRETION
   Filaments grow over time,
   freezing in earlier spacing

5. MAGNETIC FIELDS
   Additional support changes
   fragmentation scale

6. TURBULENCE
   Creates density variations
   along filaments

CONCLUSION:
Multiple effects likely combine to reduce
fragmentation wavelength from 4× to ~2×.

This is NOT an error in observations,
but a sign that theory needs refinement!
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